Lightweight Remote-Sensing Image Super-Resolution via Attention-Based Multilevel Feature Fusion Network

Hongyuan Wang, Shuli Cheng, Yongming Li, and Anyu Du

***Abstract*— In recent years, advancements in remote-sensing image super-resolution (RSISR) have achieved remarkable performance. However, many methods demand significant computational resources. This is problematic for edge devices with limited computational capabilities. To alleviate this problem, we propose an attention-based multilevel feature fusion network (AMFFN) to enhance the resolution of RSIs. This proposed network integrates three efficient design strategies to provide a lightweight solution. Initially, we designed the partial shallow residual block (PSRB) to replace the redundant convolution operation. The PSRB optimizes feature extraction via partial convolution (PConv) and capitalizes on information across channels using pointwise convolution (PWConv). Subsequently, integrating the PSRB, our dynamic feature distillation block (DFDB) leverages an information distillation mechanism to distill and capture only the crucial features securing a robust feature depiction. Conclusively, for superior feature fusion, we conceptualized an attention-based multilevel feature fusion (AMFF) mechanism. The attention intrinsic to AMFF weighs the significance of features from varied branches, assuring that the resulting output is comprehensive and discerning. We conduct thorough experimental validation on two datasets of RSIs and measure network complexity by evaluating network parameters and multiadds operations. The results show that our method effectively balances computational complexity and performance. In addition, we have expanded the application of AMFFN to the field of natural image super-resolution (SR). Experimental results on five benchmark test datasets further confirm the effectiveness of our method.**

***Index Terms*— Attention-based multilevel feature fusion (AMFF), image super-resolution (SR), lightweight neural network, remote sensing.**

I. Introduction

REMOTE-SENSING images (RSIs) have become indispensable for modern applications by providing diverse observational data. They are pivotal for various tasks, such as object detection [1], environmental surveys [2], and scene classification [3]. The detailed information from these images is vital for decision-making, resource management, and environmental protection. However, the increasing demand for high-resolution (HR) RSIs often faces challenges. For instance, the constraints of imaging equipment and adverse imaging conditions [4] frequently result in compromised image quality. Furthermore, additional factors such as transmission noise and motion blur [5] further degrade the quality of these images. RSI super-resolution (RSISR) is a technique that focuses on reconstructing a low-resolution (LR) RSI into a corresponding HR RSI. This technology not only addresses the limitations of hardware, but also enhances image resolution with reduced computational costs. As a result, RSISR has become a prominent approach for obtaining HR RSIs.

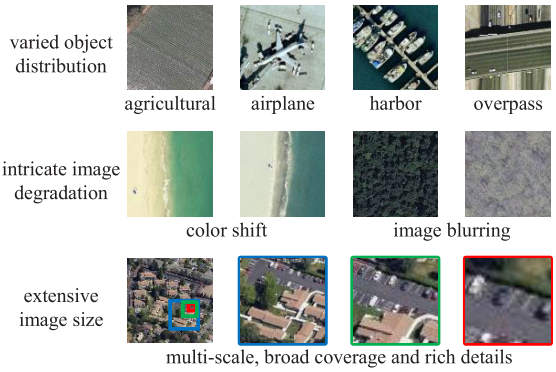


Fig. 1. Illustration of RSIs.

RSISR is not merely about producing visually appealing images; it is about unveiling details that are not visible in LR images. These details empower researchers and decision-makers to delve deeper, discerning subtle changes and patterns that might be overlooked. Furthermore, with the fusion of RGB images and hyperspectral data, RSISR can deliver a detailed representation that encompasses both spectral and spatial [6]. When brought to light, these hidden details provide a more granular and comprehensive view of the Earth's surface. As shown in Fig. 1, RSIs exhibit unique characteristics distinct from natural images, including varied object distribution, intricate image degradation, and extensive image size [7]. These attributes amplify the complexity of the RSISR. With the advancement of deep neural networks [8], [9], [10], [11], RSISR has made significant progress, giving rise to various methods such as unsupervised super-resolution (SR) [12], [13], [14] and blind SR [15], among others.

Manuscript received 13 November 2023; revised 20 November 2023; accepted 22 November 2023. Date of publication 23 November 2023; date of current version 1 December 2023. This work was supported in part by the Scientific and Technological Innovation 2030 Major Project under Grant 2022ZD0115800, in part by the Basic Research Funds for Colleges and Universities in Xinjiang Uygur Autonomous Region under Grant XJEDU2023P008, in part by the Key Laboratory Open Projects in Xinjiang Uygur Autonomous Region under Grant 2023D04028, and in part by the Natural Science Foundation Program of Xinjiang Uygur Autonomous Region under Grant 2022D01C82. *(Corresponding author: Shuli Cheng.)*

The authors are with the School of Computer Science and Technology, Xinjiang University, Ürümqi 830046, China (e-mail: why5200@stu.xju.edu.cn; cslxju@xju.edu.cn; lym@xju.edu.cn; anydxju@xju.edu.cn).

Digital Object Identifier 10.1109/TGRS.2023.3336524

1558-0644 © 2023 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 09:18:05 UTC from IEEE Xplore. Restrictions apply.

Numerous models [16], [17], [18], [19], [20], [21], [22], [23], [24] have emerged as dependable methods for RSISR, demonstrating substantial enhancements in both reconstruction accuracy and generalization.

RSISR builds upon the foundational studies of SR on natural images, introducing specific refinements for RSIs. These refinements include the formulation of loss functions that resonate with the unique characteristics of RSIs [16], the dual-stage design that leverages both spatial and spectral knowledge from adjacent bands [17], and the crafting of a degradation model that mirrors remote-sensing scenarios [18]. Zhang et al. [19] employed a hybrid higher-order attention mechanism, guiding networks toward restoring high-frequency details. Similarly, Lei et al. [20] implemented transformer-based multilevel enhancement structures to harness features at various levels. Tu et al. [21] introduced a GAN, merging the strengths of convolutional layers with the Swin transformer, to elevate the perceptual quality of restored RSIs. Although these methods have successfully reconstructed clearer RSIs, they have faced challenges in effectively managing the network's inherent complexity. The extensive parameters considerably limit their deployment on resource-constrained devices. Consequently, there is a pressing need for research on lightweight RSISR that can efficiently handle large volumes of RSIs.

Recent studies have delved into lightweight RSISR. CTN [25] designed the utilization of the contextual transformation layer (CTL) as a replacement for the 3×3 convolutional layer. This adaptation effectively reduces the network parameters. However, the design of CTL adds additional operations and connections compared to convolution, consequently leading to an increase in the inference process time. Feature enhancement network (FeNet) [26] embraced the notion of lightweight module nesting, crafting the lightweight lattice block (LLB) based on channel separation. However, the integration of weighted cross-combination in LLB introduces redundancy within the network. Meanwhile, these methods recognize the importance of feature fusion. However, there are still areas for improvement in the existing method of fusing features.

To address the limitations of prior methods, we present the attention-based multilevel feature fusion network (AMFFN) for RSISR that is both lightweight and accurate. As shown in Fig. 2, our AMFFN not only exhibits reduced network parameters, but also showcases a competitive edge over other lightweight methods in performance, achieving a better balance between network complexity and reconstruction accuracy.

Fig. 3 presents the detailed architecture of AMFFN. The core structure encompasses the dynamic feature distillation block (DFDB) and the attention-based multilevel feature fusion (AMFF). The unique construction of DFDB integrates the partial shallow residual block (PSRB) and harnesses the information distillation mechanism. A noteworthy here is the introduction of the adjustable dilation rate, amplifying its adaptability and versatility across varying scenarios. The PSRB combines partial convolution (PConv) [27] and pointwise convolution (PWConv) techniques. The former convolves only some of the channels, guiding the network to focus on important features and reducing network parameters, while the latter realizes the effective interaction of all channels. This combination allows PSRB to utilize fewer parameters while achieving a feature representation similar to that of the 3×3 convolution.

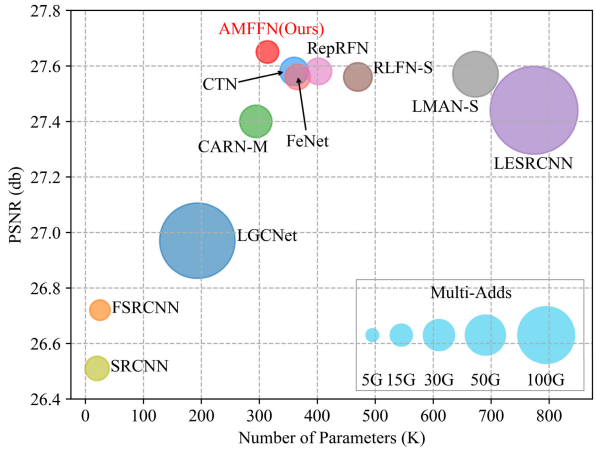


Fig. 2. Performance comparison with other lightweight methods on the UC Merced test dataset (×4). The multiadds are computed on a 1280 × 720 HR image.

Unlike other feature fusion structures [28], [29], [30], we fully use multilevel features in designing AMFF. On the one hand, we adopt a grouping method to fuse features from different blocks step by step. It enables the model to extract and fuse key features at multiple levels. On the other hand, we introduced contrast-aware channel attention (CCA) [31] to identify and enhance key features more accurately with fewer parameters. Grounded on AMFF, our network exemplifies exceptional feature fusion capabilities while maintaining a lightweight stature.

Additionally, we underscore the versatility of AMFFN by deploying it for natural image SR tasks. Our method exhibits superior performance on widely recognized benchmarks, attesting to its expansive applicability and robust capability.

Overall, the main contributions of this work are as follows.

1) We propose a lightweight RSISR network called AMFFN, which reconstructs higher-quality RSIs with fewer parameters and multiadds operations on publicly available, commonly used datasets compared to other state-of-the-art methods.
2) We propose a novel DFDB that combines PSRB and the information distillation mechanism, minimizing unnecessary redundancy in the network while extracting meaningful feature information. Meanwhile, the adjustable dilation rate enhances its adaptability and versatility.
3) We propose a novel AMFF structure to build an AMFFN that efficiently fuses key features from multilevel to obtain more accurate contextual information.

The rest of this article is organized as follows. Section II contains a comprehensive description of the related work. Section III thoroughly discusses the proposed AMFFN. Section IV presents our experimental datasets, results, and ablation studies. The conclusion is drawn in Section V.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 09:18:05 UTC from IEEE Xplore. Restrictions apply.

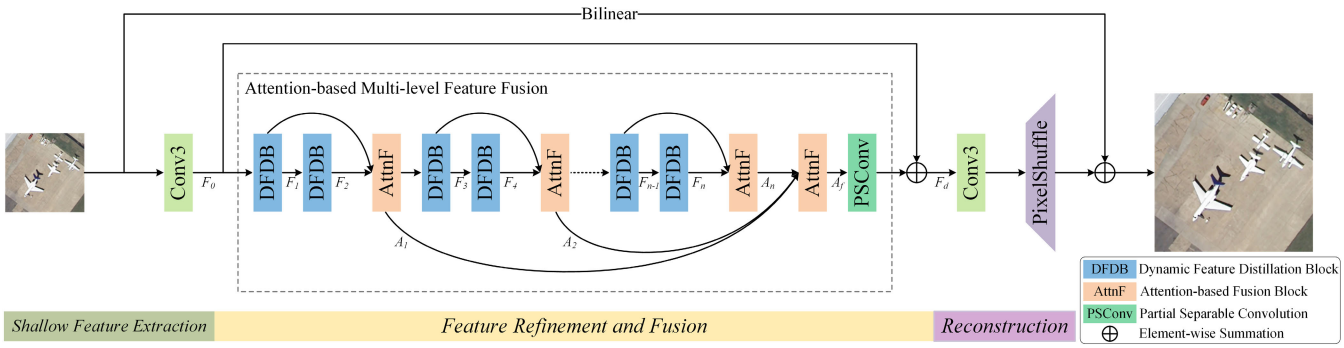


Fig. 3. Network overview. There are three primary components to our AMFFN: shallow feature extraction, feature refinement and fusion, and reconstruction. The feature refinement and fusion module enhances feature representation through the DFDB (Section III-C) and achieves a more holistic representation using the AMFF (Section III-D).

II. Related Work

A. Natural Image SR

With the renaissance of deep neural networks, numerous SR networks have been designed and have made significant progress. SRCNN [32] pioneered the application of a three-layer convolutional network in the realm of CNN-based SR, achieving a notable edge over traditional algorithms. FSRCNN [33] incorporated a deconvolutional layer for upsampling and directly utilized the original LR image as input, thus augmenting performance without adding to the complexity. Kim et al. [34] proposed a very deep convolutional network (VDSR) based on the ResNet residual structure, lowering the level of training difficulty for deep neural networks using residual learning. Taking inspiration from the squeeze-and-excite block, Zhang et al. [35] devised RCAN, an intricate network that amalgamates the channel attention module within a residual-in-residual architecture. To make optimal use of the hierarchical traits of the residual branches, Liu et al. [36] devised RFANet, a network designed to generate even more representative features.

Recently, with the introduction of the transformer model to computer vision tasks, a series of transformer-based networks have improved SR performance. Liang et al. [37] presented SwinIR based on the Swin transformer, which promotes the Swin transformer for the SR task. Chen et al. [38] proposed a novel hybrid attention transformer (HAT) to activate more input pixels for reconstruction.

B. SR for RSIs

Deep learning-based SR networks have demonstrated remarkable performance in enhancing the quality of natural images. Inspired by this success, many researchers have endeavored to design SR networks that are specifically optimized for RSIs. Lei et al. [39] presented LGCNet, an RSISR network leveraging CNNs. This network enhances the ability to learn from spatial data by integrating information from multiple feature maps. Haut et al. [40] unveiled a deep compendium network (DCM) that harnesses residual units and skip connections to capture salient features across varied image regions. In a quest to refine the reconstruction of large-scale RSIs, Dong et al. [41] developed a dense sampling SR network (DSSR). Meanwhile, Lei et al. [20] presented TransENet, where the use of transformers is introduced to take advantage of features at various levels. Wang et al. [26] constructed a lightweight FeNet aiming to strike an optimal equilibrium between computational cost and reconstruction accuracy.

Moreover, many researchers have investigated continuous-scale SR to address the challenge of fixed scale factors. Wu et al. [42] proposed a scale-aware dynamic network (SADN) tailored for continuous-scale SR. Similarly, Chen et al. [43] advanced a versatile continuous-scale SR framework, termed FunSR, that leverages context interaction within the implicit function space.

C. Lightweight Image SR

The quest for enhanced performance in current SR networks often results in significant computational overhead, posing challenges for practical deployment. Recognizing this, there has been a discernible shift toward the development of efficient and lightweight image SR, especially as the demand for lightweight SR network deployment grows amid constrained computational resources. Ahn et al. [44] introduced the cascading residual network (CARN) that emphasizes network lightweight. Hui et al. [45] developed the information distillation network (IDN), which amalgamates diverse features to distill salient information. Building on this, IMDN [31] augmented IDN using an information multidistillation block and channel splitting operation. Liu et al. [46] revisited the channel splitting technique and introduced the progressive refinement module as an alternative. They proposed a lightweight and precise residual feature distillation network (RFDN). Recently, Kong et al. [47] designed RLFN, ditching the multibranch design of RFDN, and introduced a contrastive loss, resulting in expedited computations and heightened reconstruction accuracy.

In contrast to methods that primarily focus on developing lightweight modules, our method prioritizes a more lightweight feature fusion structure and focuses on reducing the burden on the convolutional layer. We adopt strategies that facilitate a comprehensive feature representation using fewer parameters and multiadds operations.

III. Proposed Method

In this section, we describe the architecture of the proposed AMFFN network and define the loss function employed for its

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 09:18:05 UTC from IEEE Xplore. Restrictions apply.

optimization. Subsequently, we provide a detailed exposition of the PSRB and the DFDB. We then introduce the AMFF. Lastly, we offer insightful discussions pertaining to AMFFN.

A. Network Architecture

In this work, we propose an innovative network, AMFFN, designed for lightweight RSISR. Fig. 3 depicts the network architecture, comprising three integrated components: shallow feature extraction $\mathcal{H}_{\text{fe}}$, feature refinement and fusion $\mathcal{H}_{\text{rf}}$, and reconstruction $\mathcal{H}^s_{\text{re}}$. Let the input and output of AMFFN be represented as $I_{\text{LR}} \in \mathbb{R}^{H\times W\times 3}$ and $I_{\text{SR}} \in \mathbb{R}^{sH\times sW\times 3}$, respectively, where $H \times W$ indicates the spatial dimensions and s is the upsampling factor. The process to derive the output for our model proceeds as follows:

$$F_0 = \mathcal{H}_{\text{fe}}(I_{\text{LR}}) \tag{1}$$

$$F_d = \mathcal{H}_{\text{rf}}(F_0) \tag{2}$$

$$I_{\text{SR}} = \mathcal{H}^s_{\text{re}}(F_d) + \mathcal{H}^s_{\text{bilinear}}(I_{\text{LR}}) \tag{3}$$

where F_0 and F_d represent the feature representations from $\mathcal{H}_{\text{fe}}$ and $\mathcal{H}_{\text{rf}}$. We utilize a 3×3 convolutional layer to implement $\mathcal{H}_{\text{fe}}$, which extracts the initial representation of the I_{LR} content. Similarly, $\mathcal{H}_{\text{re}}$ is realized through a 3×3 convolutional layer combined with a pixelshuffle layer. Notably, $\mathcal{H}^s_{\text{bilinear}}$ denotes the bilinear interpolation function that upsamples by a factor of s. This function, as a mathematical computation method, possesses the merit of being parameter-free. It effectively relays vast information, guiding the network's focus toward intricate detail reduction.

To develop a lightweight network for RSISR, we meticulously designed the feature refinement and fusion module $\mathcal{H}_{\text{rf}}$. This module achieves a profound feature representation using an efficient approach. Specifically, it incorporates two essential components: the DFDB (Section III-C) and the AMFF (Section III-D).

We trained the aforementioned network using the L_1 loss function. For a training set comprising N pairs of LR and HR images, represented by $\{I^i_{\text{LR}}, I^i_{\text{HR}}\}^N_{i=1}$, the loss is determined as follows:

$$L_1(\boldsymbol{\theta}) = \frac{1}{N}\sum_{i=1}^{N}\left\| I^i_{\text{SR}} - I^i_{\text{HR}} \right\|_1 \tag{4}$$

where θ denotes the parameter sets of our proposed AMFFN.

B. Partial Shallow Residual Block

In the realm of neural network optimization, the PConv [27] mechanism presents an effective strategy to minimize computational overhead while preserving the depth of feature extraction. The PConv operates by selectively applying the convolution to a designated subset of input channels, effectively concentrating computational resources where they are most needed. Fig. 4(a) graphically illustrates this mechanism, depicting the convolution operation as limited to a portion of the input channels, thereby delineating a concentrated and efficient modality for extracting features.

Inspired by the PConv, we design a novel PSRB as a lightweight alternative to the ordinary convolution. The block

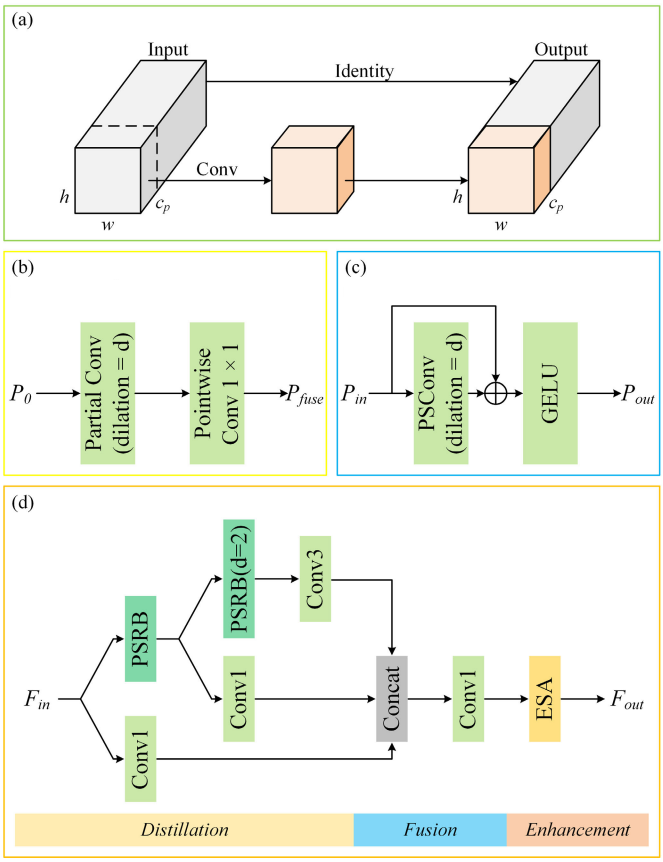


Fig. 4. Network structures of essential modules. ⊕ represents element-wise summation. (a) PConv. (b) PSConv. (c) PSRB. (d) DFDB.

is shown in Fig. 4(c), with GELU [48] chosen as the activation function. The structure comprising PConv and PWConv is referred to as partially separable convolution (PSConv), as seen in Fig. 4(b).

Given an input feature $P_{\text{in}} \in \mathbb{R}^{H\times W\times C}$, the PSRB initiates the transformation process through PConv, which, unlike conventional convolutions that engage the entire channel set, selectively convolves only a fraction, thereby drastically reducing the computational cost. This is mathematically formalized as

$$P_0 = \mathcal{H}_{\text{PConv}}(P_{\text{in}}, c_p, d) \tag{5}$$

where P_0 represents the output after the PConv, c_p represents the selected channels, and d indicates the dilation rate. The default value of the parameter c_p is one-third of C and the parameter d defaults to 1.

The PConv ensures that the computation is reduced to the original $(c_p/C)^2$, a significant decrease from the traditional convolution, exemplifying a direct correlation between the reduced channel engagement and the lowered computational demand. The dilation rate is introduced to expand the receptive field without increasing network parameters, effectively enhancing the field of view over the input.

Subsequently, the PSRB utilizes a PWConv to process the output P_0, integrating interactions across the channel dimensions and yielding an intermediate state P_{fuse}

$$P_{\text{fuse}} = \mathcal{H}_{\text{PWConv}}(P_0) \tag{6}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 09:18:05 UTC from IEEE Xplore. Restrictions apply.

where P_{fuse} encapsulates the fused information from the PConv. This step is crucial in maintaining the integrity of the feature representation, assuring that despite the reduced computational expense, the network's depth of processing remains uncompromised.

The ending of the process about PSRB is the application of the GELU activation function to the summative combination of the input P_{in} and the fused output P_{fuse}, resulting in the final output P_{out}

$$P_{\text{out}} = \mathcal{H}_{\text{GELU}}(P_{\text{in}} + P_{\text{fuse}}). \tag{7}$$

Through this sophisticated sequence of operations, the PSRB outputs enhanced feature representations, capitalizing on the efficiency of PConv while maintaining the richness of the input data.

C. Dynamic Feature Distillation Block

Receptive field diversity is essential for capturing varying contextual information, which is particularly critical in the reconstruction of RSIs. Drawing on the principles of information distillation, we have conceptualized a novel DFDB, as illustrated in Fig. 4(d). This block is structured to undergo three distinct phases: feature distillation, feature fusion, and feature enhancement, each contributing to the progressive refinement of features.

The distillation phase employs the PSRB, along with additional convolutional layers to progressively refine the input features F_{in} and distill key features. Coarse features are those that have been initially refined but still require further processing. Distilled features are the key features extracted through the distillation process. The distillation process halves the number of feature channels, thereby selectively extracting key features. This is formulated as follows:

$$\begin{cases} F_1^{\text{coarse}} = \mathcal{H}_{\text{PSRB}}(F_{\text{in}}) \\ F_2^{\text{coarse}} = \mathcal{H}_{\text{PSRB}}\left(F_1^{\text{coarse}}, d = 2\right) \\ F_1^{\text{distilled}} = \text{Conv1}(F_{\text{in}}) \\ F_2^{\text{distilled}} = \text{Conv1}\left(F_1^{\text{coarse}}\right) \\ F_3^{\text{distilled}} = \text{Conv3}\left(F_2^{\text{coarse}}\right) \end{cases} \tag{8}$$

where F_n^{coarse} and $F_n^{\text{distilled}}$ denote the nth coarse and distilled features, respectively. $\mathcal{H}_{\text{PSRB}}$, indicating the PSRB, is responsible for the refinement process to enhance feature representation. Meanwhile, the Conv1 and Conv3 layers are responsible for the distillation process, focusing on the extraction of key features.

In the fusion phase, the distilled features from different layers are concatenated and subsequently processed by a 1×1 convolutional layer to synthesize the fused feature F_{fusion}, described by

$$F_{\text{fusion}} = \text{Conv1}\left(\text{Concat}\left(F_1^{\text{distilled}}, \ldots, F_3^{\text{distilled}}\right)\right) \tag{9}$$

where F_{fusion} represents the fused feature.

Finally, the enhancement phase introduces an attention mechanism to amplify the spatial details within the fused features, described by

$$F_{\text{out}} = \mathcal{H}_{\text{ESA}}(F_{\text{fusion}}) \tag{10}$$

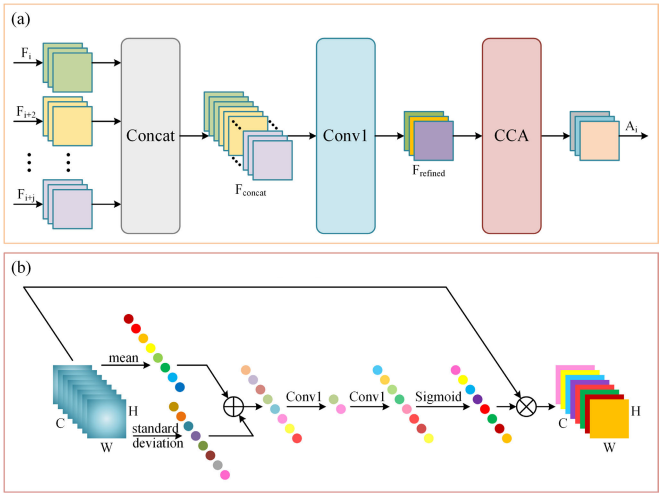


Fig. 5. Illustration of AttnF. $\oplus$ and $\otimes$ represent element-wise summation and element-wise multiplication, respectively. (a) AttnF. (b) CCA.

where F_{out} is the enhanced feature and $\mathcal{H}_{\text{ESA}}$ denotes the enhanced spatial attention (ESA) [36] block that effectively enhances the model capability from a spatial perspective. The number of convolutional layers in ConvGroups in ESA to one. Benefiting from the ESA, DFDB can effectively acquire adequate contextual information, aiding the network in producing more accurate visual outcomes.

D. Attention-Based Multilevel Feature Fusion

In the domain of RSISR, the exploitation of hierarchical information is paramount. To this end, we develop an AMFF structure. Our designed AMFF structure can integrate feature information at various levels, capturing low-level details and high-level semantics for an enhanced representation.

The core of our feature fusion structure is the attention-based fusion block (AttnF), shown in Fig. 5(a). It begins with the concatenation of the n input feature, each having C channels. This process can be expressed as

$$F_{\text{concat}} = \text{Concat}(F_i, F_{i+1}, \ldots, F_{i+n}) \tag{11}$$

where F_{concat} is the concatenated feature and F_{i+n} is the feature of the nth input. Then, the dimensionality reduction is performed by 1×1 convolutional layer, described by

$$F_{\text{refined}} = \text{Conv1}(F_{\text{concat}}) \tag{12}$$

where F_{refined} is the refined feature, reducing the channel dimension back to C, retaining the most salient features.

F_{refined} is then enhanced by the CCA [31], which emphasizes the most informative features. The specific formula is expressed as follows:

$$A_i = \mathcal{H}_{\text{CCA}}(F_{\text{refined}}) \tag{13}$$

where A_i denotes the final output of AttnF. Here, we briefly describe the calculation of the CCA used in AttnF. As shown in Fig. 5(b), given a group of feature maps, we calculate the summation of their standard deviation and mean. This summation is then passed to a nonlinear function sequence: Conv1 $\rightarrow$ Relu $\rightarrow$ Conv1. Subsequently, a sigmoid function is

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 09:18:05 UTC from IEEE Xplore. Restrictions apply.

utilized to generate the combination coefficients, which range from 0 to 1. Finally, we multiply the input feature maps with these combination coefficients to obtain the output feature maps. This construction allows the AttnF to adapt to different channel dimensions and enhances the feature representation.

The AMFF structure we have devised comprises three distinct operation phases. In the first phase, we designate the ith DFDB output as F_i and the ith AttnF module as A_i. To embark on the fusion process, the fusion of different levels of DFDB features in a grouped form using AttnF results in a seamless and continuous refinement and expansion of these features. Denoting these features as $F_i(i = 1, \ldots, 8)$ and $A_i(i = 1, \ldots, 4)$ in ascending index order from left to right in Fig. 3. The fusion operation at the first phase can then be expressed as

$$A_i = \mathcal{H}_{\text{AttnF}}(F_{2i-1}, F_{2i}), \quad i = 1, \ldots, 4 \tag{14}$$

where $\mathcal{H}_{\text{AttnF}}$ is the AttnF block. For the second phase, we add further refinement to the results. The formula for the second phase fusion operation is

$$A_f = \mathcal{H}_{\text{AttnF}}(A_1, \ldots, A_4) \tag{15}$$

where A_f is the condensed feature, and $\mathcal{H}_{\text{AttnF}}$ is the AttnF block. Finally, adding a convolutional layer after using the fusion structure to deepen the features has been widely proven effective. Therefore, we finish by adding a PSConv to further enhance the features. The formula for the third phase fusion operation is

$$F_{\text{out}} = \mathcal{H}_{\text{PSConv}}(A_f) \tag{16}$$

where F_{out} is the fusion feature and $\mathcal{H}_{\text{PSConv}}$ is the PSConv. With the AMFF structure, the interaction between feature refinement and fusion is performed iteratively, gradually refining and enriching the features. More significant contextual information can be extracted, thanks to it.

E. Discussion

1) Why Does AMFFN Have Such a Small Number of Parameters and Multiadds Operations?: The architecture of AMFFN significantly reduces the number of parameters and multiadds operations while maintaining excellent reconstruction accuracy through three design elements. The PSRB combines partial and pointwise convolutions, reducing the redundant convolution operation while maintaining effective feature extraction. The DFDB employs an information distillation mechanism, enabling gradual refinement of the complex information derived from PSRB, which leads to an optimal feature representation while reducing model complexity. The AMFF meticulously extracts and integrates essential features across multiple levels, enhancing the network's capacity for detail where only a small amount of complexity is added. These innovative components collectively enable our network to deliver excellent performance with less network complexity.

2) Difference From Other Lightweight SR Methods?:

a) Partial shallow residual block: The PSRB has been developed as a more efficient alternative to traditional convolution, with the aim of enhancing feature extraction efficiency and reducing redundancy. This is achieved through the integration of partial and pointwise convolutions, which collectively obviate the need for superfluous computations. The implementation of the PSRB confers multiple benefits: it necessitates fewer parameters and multiadds operations, and it adeptly captures contextual information across different scales. Consequently, this facilitates the extraction of more comprehensive and informative features.

b) Dynamic feature distillation block: The information distillation mechanism has significantly advanced the development of lightweight SR networks and is increasingly integrated into lightweight RSISR networks. In contrast to the distillation blocks in networks such as IMDN and RFDN, our DFDB enables an equivalence feature representation by leveraging shorter connection paths during the distillation process. Unlike its counterparts, which typically undergo four stages of distillation, DFDB accomplishes comparable outcomes with only three stages. Additionally, DFDB's integration of the adjustable dilation rate augments the network's receptive field, bolstering its adaptability and generalizability.

c) Attention-based multilevel feature fusion: As depicted in Fig. 6, our novel structure contrasts with existing fusion structures [28], [29], [30] by aiming to enhance performance without compromising the model's complexity. To accomplish this, we rethought the standard feature fusion. Traditional channel concatenation invariably leads to excessive stacking of useless features, whereas a 1×1 convolutional layer often fails to adequately emphasize key features. To address these challenges, we implemented two key modifications: first, we applied a grouping method to the fusion of block features, and second, we incorporated an attention-based fusion block. This innovative configuration culminates in an efficient and comprehensive multilevel feature fusion. For an extensive comparison with other fusion structures, the reader is referred to Section IV-C.

3) Why Does AMFFN Perform Well For RSISR?: Compared to natural images, RSIs display distinct scenes, each of which usually encompasses a complex distribution of targets. Furthermore, targets within these images are generally of a smaller scale relative to those in natural imagery, posing significant challenges to RSISR. The AMFFN we propose adeptly confronts these obstacles through meticulous local and global processing. Locally, it harnesses the capabilities of both the PSRB and DFDB for effective feature extraction and enhancement, safeguarding the detail fidelity of all targets, regardless of size. Globally, the AMFF framework integrates disparate levels of information, facilitating the creation of high-fidelity RSIs. Illustrated in Fig. 7, the visualization of average feature maps unveils the AMFFN's methodical approach: the DFDB refines features in increasing depth, bringing forth a richer detail profile. Concurrently, AttnF serves as a discerning filter, directing the model's focus toward the most pertinent information, ensuring that the resultant images are coherent on a macrolevel and precise at a microlevel. AMFFN demands fewer parameters and multiadds operations, providing a viable strategy for improving spatial resolution without compromising computational efficiency. The experimental results in Section IV substantiate these advantages and confirm the superiority of our method.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 09:18:05 UTC from IEEE Xplore. Restrictions apply.

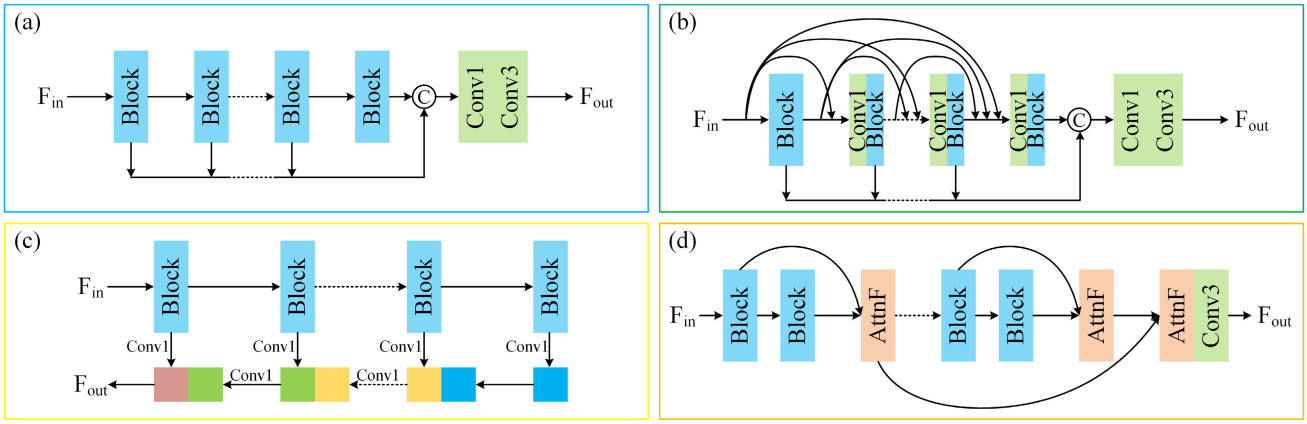


Fig. 6. Comparisons of different feature fusion structures. (a) Basic unit of the standard feature fusion. (b) Structure of the dense feature fusion. (c) Structure of the backward fusion module. (d) Structure of the AMFF. © represents concatenation. The block denotes the backbone block in the network and the AttnF denotes the attention-based fusion block.

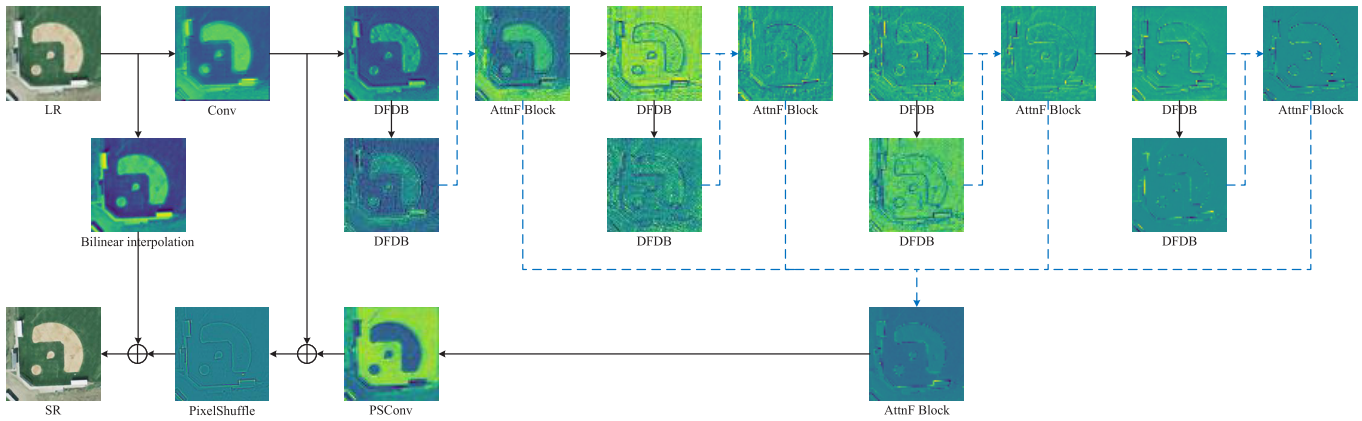


Fig. 7. Network average feature maps visualization. ⊕ represents element-wise summation. The DFDB denotes the dynamic feature distillation block, the AttnF denotes the attention-based fusion block, and the PSConv denotes partial separable convolution.

IV. Experimental Results and Analyses

In this section, the performance of the AMFFN is benchmarked against leading-edge methods for RSISR, as elaborated in Section IV-A. Subsequently, the utility of AMFFN for SR of natural images is evaluated, with pertinent details and findings presented in Section IV-B. For each domain, we present dedicated discussions encompassing datasets, implementation specifics, and empirical outcomes. Comprehensive ablation studies are detailed in Section IV-C to deepen the understanding of our network's architecture and functionality.

A. Main Task: RSISR

1) Datasets: For our comparative analyses, we utilized two widely recognized public datasets of RSIs: the UC Merced dataset[1] [49] and the RSSCN7 dataset[2] [50]. The UC Merced dataset contains 2100 images, each 256 × 256 pixels, covering 21 diverse scene categories. The RSSCN7 dataset includes 2800 images, with a resolution of 400 × 400 pixels, across seven scene categories. Following [5] and [20], we split each dataset into two halves for training and testing, where 20% of the training set is taken as validation.

Each dataset's original image is regarded as a genuine HR reference, with bicubic interpolation employed to generate the corresponding LR image. All images have been converted to the PNG format for integration into the SR models, facilitating the creation of HR and LR image pairs essential for training and evaluation.

2) Metrics: We utilize the peak signal-to-noise ratio (PSNR) and the structural similarity index (SSIM) [51] as our primary evaluation metrics for comparative analysis, consistent with numerous preceding works. We calculated the PSNR and the SSIM in the RGB color space and assessed the model's complexity by examining the number of parameters and multiadds operations.

3) Implementation Details: Consistent with many preceding methods in RSISR, our approach initially employs data augmentation to capitalize fully on the available training data. This includes applying random rotation and horizontal flips to enhance the dataset. We employ the Adam optimizer to train our neural network, adopting a batch size of 16. During the training phase, we randomly select fixed-size patches from

[1]All these 21 classes: 1) Agricultural; 2) Airplane; 3) Baseball diamond; 4) Beach; 5) Buildings; 6) Chaparral; 7) Dense residential; 8) Forest; 9) Freeway; 10) Golf course; 11) Harbor; 12) Intersection; 13) Medium residential; 14) Mobile home park; 15) Overpass; 16) Parking lot; 17) River; 18) Runway; 19) Sparse residential; 20) Storage tanks; and 21) Tennis court.

[2]All these seven classes: 1) Grass; 2) Field; 3) Industry; 4) River Lake; 5) Forest; 6) Resident; and 7) Parking.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 09:18:05 UTC from IEEE Xplore. Restrictions apply.

TABLE I

QUANTITATIVE RESULTS FOR SR ON TWO RSI TEST DATASETS ARE PRESENTED. RESULTS THAT ARE NOT AVAILABLE ARE INDICATED WITH A "-". THE BEST RESULTS ARE IN BOLD. THE MULTIADDS IS COMPUTED ON A 1280 × 720 HR IMAGE

Method	Scale	Params	M-Adds	UC Merced PSNR/SSIM	RSSCN7 PSNR/SSIM	Scale	Params	M-Adds	UC Merced PSNR/SSIM	RSSCN7 PSNR/SSIM
Bicubic		-	-	30.54/0.8747	30.43/0.8446		-	-	27.18/0.7587	27.93/0.7306
SRCNN [32]		20K	18.4G	32.80/0.9144	31.91/0.8873		20K	18.4G	28.68/0.8091	28.85/0.7764
FSRCNN [33]		25K	15.1G	33.00/0.9167	32.00/0.8890		25K	13.7G	28.87/0.8158	28.96/0.7816
LGCNet [39]		193K	177.9G	33.61/0.9240	32.30/0.8944		193K	178.1G	29.28/0.8276	29.16/0.7885
CARN-M [44]		257K	91.2G	34.01/0.9277	32.43/0.8969		303K	46.2G	29.73/0.8400	29.33/0.7943
LESRCNN [52]		626K	281.3G	34.00/0.9278	32.41/0.8964		811K	239.4G	29.74/0.8405	29.34/0.7938
CFSRCNN [53]	×2	1310K	464.4G	34.14/0.9291	32.51/0.8981	×3	1495K	339.7G	29.84/0.8430	29.41/0.7975
LMAN-S [54]		525K	119.9G	**34.19/0.9295**	32.54/0.8992		710K	73.1G	29.97/0.8461	29.43/0.7975
CTN [25]		349K	58.9G	34.17/0.9292	32.53/0.8987		349K	36.6G	29.94/0.8450	29.42/0.7977
FeNet [26]		351K	77.2G	34.18/0.9292	32.53/0.8986		357K	35.0G	29.95/0.8448	29.42/0.7972
RLFN-S [47]		455K	98.8G	34.15/0.9292	32.55/0.8992		461K	44.6G	29.95/0.8456	29.43/0.7978
RepRFN [55]		386K	84.9G	34.18/0.9291	32.52/0.8986		392K	38.4G	29.96/0.8453	29.43/0.7976
AMFFN(Ours)		298K	60.2G	34.18/0.9292	**32.56/0.8993**		305K	27.4G	**30.01/0.8467**	**29.45/0.7985**
Bicubic		-	-	25.33/0.6651	26.61/0.6498		-	-	22.21/0.4885	24.36/0.5135
SRCNN [32]		20K	18.4G	26.51/0.7184	27.32/0.6937		20K	18.4G	22.71/0.5203	24.71/0.5351
FSRCNN [33]		25K	13.2G	26.72/0.7256	27.45/0.6994		25K	12.7G	22.93/0.5289	24.85/0.5402
LGCNet [39]		193K	177.9G	26.97/0.7365	27.57/0.7056		193K	177.9G	22.97/0.5344	24.88/0.5442
CARN-M [44]		294K	32.5G	27.40/0.7543	27.79/0.7156		331K	17.8G	23.33/0.5579	25.10/0.5569
LESRCNN [52]		774K	241.4G	27.44/0.7573	27.79/0.7161		922K	231.4G	23.31/0.5567	25.11/0.5572
CFSRCNN [53]	×4	1458K	312.6G	27.52/0.7601	27.83/0.7180	×8	1606K	274.7G	23.31/0.5558	25.12/0.5588
LMAN-S [54]		673K	65.1G	27.57/0.7612	27.85/0.7185		820K	51.5G	23.44/**0.5621**	**25.17**/0.5596
CTN [25]		360K	25.2G	27.58/0.7622	27.87/0.7193		371K	16.8G	23.28/0.5590	25.07/0.5548
FeNet [26]		366K	20.2G	27.56/0.7612	27.85/0.7180		428K	5.9G	23.42/0.5592	25.16/0.5580
RLFN-S [47]		470K	25.6G	27.56/0.7610	27.84/0.7184		533K	7.3G	23.41/0.5571	25.06/0.5502
RepRFN [55]		402K	22.1G	27.58/0.7611	27.85/0.7183		464K	6.4G	23.19/0.5410	24.98/0.5462
AMFFN(Ours)		314K	15.9G	**27.65/0.7641**	**27.88/0.7200**		376K	4.9G	**23.45**/0.5613	**25.17/0.5602**

LR images to serve as inputs for the network. Precisely, for upsampling factors of ×2, ×3, and ×4, the extracted patches measure 48 × 48 pixels, while for an ×8 upsampling factor, the patch size is reduced to 24 × 24 pixels. The initial learning rate is 5×10^{-4}. The total training epochs is 2000, and the learning rate will decrease by half at 1500. We conducted all experiments within the PyTorch framework, utilizing an NVIDIA RTX 3090Ti GPU.

4) Quantitative Results: Our comparative method consists of three parts: single-image SR methods (i.e., SRCNN [32], FSRCNN [33], and CFSRCNN [53]), RSISR methods (i.e., LGCNet [39], CTN [25], and FeNet [26]), and lightweight SR methods (i.e., CARN-M [44], LESRCNN [52], LMAN-S [54], RLFN-S [47], and RepRFN [55]). The reconstruction accuracy of our AMFFN is delineated in Table I, showing that our AMFFN outperforms or is on par with existing methods. For an upsampling factor of ×2, LMAN-S, with approximately 525 K parameters, delivers better or comparable results on two RSI datasets. In comparison, our AMFFN achieves comparable performance levels with about 298 K parameters and fewer multiadds operations. The performance of AMFFN is comparable to other methods with more parameters. For an upsampling factor of ×8, it is observed that methods such as CTN and RepRFN experience a noticeable decline in performance. In contrast, our AMFFN remarkably maintains its advanced performance capabilities. This can show that our AMFFN has better advantages in robustness and stability. These attributes are crucial in high-magnification contexts, where accuracy and effectiveness are often challenging. Additionally, Tables II and III summarize the PSNR of ×4 SR for each class on the UC Merced and RSSCN7 test datasets.

TABLE II

UC MERCED TEST DATASET AVERAGE PSNR (dB) FOR EACH CLASS FOR AN UPSAMPLING FACTOR OF 4

Class	LMAN-S [54]	CTN [25]	FeNet [26]	RLFN-S [47]	RepRFN [55]	AMFFN (Ours)
1	26.56	26.82	**27.03**	26.75	26.59	26.94
2	28.85	28.85	28.76	28.80	28.84	**28.91**
3	31.65	31.62	31.64	31.67	31.65	**31.70**
4	34.76	**34.78**	34.74	34.76	34.74	34.77
5	25.86	25.90	25.82	25.79	25.85	**25.93**
6	**26.03**	25.95	25.99	26.00	26.02	**26.03**
7	26.31	26.38	26.26	26.28	26.34	**26.39**
8	26.81	26.80	26.81	26.81	26.81	**26.82**
9	28.63	28.63	28.57	28.60	**28.68**	28.65
10	**32.38**	32.37	32.35	32.35	32.35	**32.38**
11	23.49	23.45	23.46	23.39	23.52	**23.67**
12	26.19	26.23	26.21	26.23	26.27	**26.28**
13	25.15	25.17	25.11	25.11	25.14	**25.20**
14	25.12	25.20	25.09	25.02	25.11	**25.25**
15	26.68	26.53	26.60	26.62	26.67	**26.71**
16	22.78	22.84	22.80	22.88	22.98	**22.99**
17	26.96	26.96	26.96	26.95	26.96	**26.98**
18	31.49	31.42	31.48	31.59	31.57	**31.66**
19	27.90	27.91	27.91	27.89	27.91	**27.94**
20	26.94	26.94	26.90	26.91	26.90	**26.97**
21	28.38	28.35	28.33	28.41	28.39	**28.44**
AVG	27.57	27.58	27.56	27.56	27.58	**27.65**

The results indicate that AMFFN outperforms other methods in most classes, which corroborates the superior effectiveness of our proposed method.

5) Qualitative Results: The visual outcomes of the UC Merced and RSSCN7 test datasets are shown in Fig. 8 after being four times enlarged. Compared to alternative methods, the AMFFN can produce visually appealing results

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 09:18:05 UTC from IEEE Xplore. Restrictions apply.



Fig. 8. Result comparison of our method with other lightweight methods on the UC Merced and RSSCN7 test datasets (×4).

TABLE III

RSSCN7 TEST DATASET AVERAGE PSNR (dB) FOR EACH CLASS FOR AN UPSAMPLING FACTOR OF 4

Class	LMAN-S [54]	CTN [25]	FeNet [26]	RLFN-S [47]	RepRFN [55]	AMFFN (Ours)
1	**32.86**	32.85	**32.86**	32.83	32.85	32.85
2	32.23	32.23	32.24	32.21	32.23	**32.26**
3	24.82	24.85	24.81	24.80	24.81	**24.86**
4	29.85	29.87	29.84	29.86	29.85	**29.88**
5	27.10	**27.11**	27.10	27.10	27.09	**27.11**
6	24.04	24.06	24.02	24.03	24.02	**24.07**
7	24.08	24.10	24.07	24.08	24.08	**24.12**
AVG	27.85	27.87	27.85	27.84	27.85	**27.88**

TABLE IV

QUANTITATIVE COMPARISON WITH OTHER LIGHTWEIGHT RSISR METHODS ON THE UC MERCED TEST DATASET

Method	Scale	Params	M-Adds	Time	PSNR
CTN [25]	×2	349K	**58.9G**	0.0370s	34.17
FeNet [26]		351K	77.2G	0.0303s	**34.18**
AMFFN(Ours)		**298K**	60.2G	**0.0247s**	**34.18**
CTN [25]	×3	349K	36.6G	0.0338s	29.94
FeNet [26]		357K	35.0G	0.0253s	29.95
AMFFN(Ours)		**305K**	**27.4G**	**0.0210s**	**30.01**
CTN [25]	×4	360K	25.2G	0.0335s	27.58
FeNet [26]		366K	20.2G	0.0254s	27.56
AMFFN(Ours)		**314K**	**15.9G**	**0.0204s**	**27.65**
CTN [25]	×8	**371K**	16.8G	0.0348s	23.28
FeNet [26]		428K	5.9G	0.0250s	23.42
AMFFN(Ours)		376K	**4.9G**	**0.0206s**	**23.45**

with sharper edges, contours, and a better visual perception. This observation offers empirical evidence to substantiate the effectiveness of our AMFFN. For the image "storagetanks99," we observe that the majority of comparison methods result in the production of noticeable artifacts and blurring effects. Unlike them, our method can achieve reconstruction with fewer artifacts, yielding a higher-quality reconstructed image.

6) Inference Speed and Network Complexity: We evaluated the inference speed of lightweight RSISR methods on the UC Merced test dataset. The identical experimental platform was used for all network evaluations to ensure fair comparisons. Detailed comparisons outlined in Table IV illustrate

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 09:18:05 UTC from IEEE Xplore. Restrictions apply.

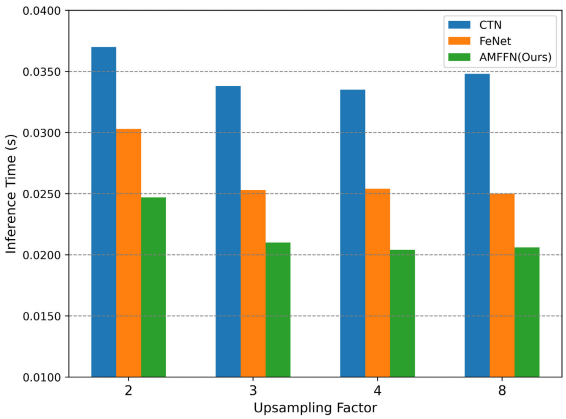


Fig. 9. Inference time by the method and the upsampling factor.

that our AMFFN not only optimizes the tradeoff between computational complexity and reconstruction accuracy, but also excels in inference speed relative to existing lightweight RSISR methods. The superiority of AMFFN in inference speed is more effectively conveyed through the bar charts in Fig. 9. The balance between these factors ensures that our method is highly adaptable to remote-sensing equipment and offers greater flexibility in its utilization.

B. Additional Task: Natural Image SR

1) Datasets: We are using the DIV2K dataset [56], containing 800 high-quality images as the training dataset. The commonly used SR benchmarks (e.g., Set5 [57], Set14 [58], B100 [59], Urban100 [60], and Manga109 [61]) are chosen for testing. Unlike RSI datasets, the DIV2K dataset consists of a variety of natural images. By extending our experiments to include the DIV2K dataset, we test and refine our models in a broader context. In line with current approaches, we employed bicubic interpolation in MATLAB to downsample the HR ground-truth images, thus creating LR counterparts at downsampling factors of $\times 2$, $\times 3$, and $\times 4$.

2) Metrics: Consistent with numerous prior studies, we have chosen the PSNR and the SSIM [51] as the evaluation metrics for image SR. All SR results are converted to YCbCr color space and evaluated on the luminance (Y) channel to ensure consistency in quality assessment. Additionally, we employ the number of parameters and multiadds operations as measures to evaluate the computational complexity of the networks.

3) Implementation Details: Consistent with many preceding methods, we use random rotation and horizontal flipping for data augmentation. We employ the Adam optimizer to train our neural network. In the training phase, we randomly crop 64×64 patches from LR images as the input of networks. The total training iterations are set at 1000k with a batch size of 64. The initial learning rate is initialized as $1e^{-3}$ and halved at [100k, 500k, 800k, 900k, 950k]-step. We conducted all experiments within the PyTorch framework, utilizing an NVIDIA RTX 3090Ti GPU.

4) Quantitative Results: We compared our proposed AMFFN with many cutting-edge lightweight SISR methods, namely IDN [45], CARN [44], IMDN [31], RFDN [46], LMAN-S [54], SMSR [62], FDIWN-M [63], VLESR [64], GASSL-S [65], and FDSCSR-S [66]. Table V lists the outcomes of this comparison. Our findings indicate that AMFFN on most testing sets is better or comparable to the other lightweight SR networks regarding reconstruction accuracy for $\times 2$, $\times 3$, and $\times 4$ SR tasks. Additionally, AMFFN achieved these superior results while utilizing fewer parameters and multiadds operations. This finding provides empirical evidence to substantiate the effectiveness of our method in achieving a favorable balance between network parameters, multiadds operations, and the accuracy of image reconstruction.

5) Qualitative Results: Fig. 10 offers a visual comparison of various SR networks on the Urban100 dataset for $\times 4$ SR. Our AMFFN demonstrates a distinct capability to render textures and edges with enhanced clarity, which further corroborates the superior performance of our approach. Notably, in the images labeled "img062" and "img096," many comparative methods exhibit noticeable distortions and blurring, whereas our AMFFN produces more precise lines. According to the analysis displayed above, our AMFFN is competitive in both natural and RSI SR.

C. Ablation Study

1) Efficacy of the Dynamic Feature Distillation Block: To assess the significance of the DFDB, we construct basic blocks that replace the feature distillation phase of DFDB with three convolutional layers and use them as the main building blocks within the network. The results of this assessment are detailed in Table VI. An analysis of the first two rows of Table VI shows that the integration of DFDB into the network not only contributes to a more lightweight architecture, but also improves reconstruction accuracy. The network parameters have been reduced to 313 K, resulting in an increase of 0.03 dB in PSNR and 0.0004 in SSIM. These improvements effectively highlight the effectiveness and superiority of incorporating DFDB.

To more comprehensively demonstrate the impact of DFDB, we present a visual analysis using the local attribution map (LAM) [67]. Fig. 11 illustrates that the integration of DFDB results in an increase in the diffusion index (DI) value from 4.59 to 6.54. The DI metric reflects the extent of pixel involvement, where a higher DI signifies a broader range of attention. Furthermore, the enhanced SR outcomes further underscore the efficacy of DFDB. These advancements collectively underscore the superiority of DFDB.

We thoroughly examined the optimal hyperparameter settings for the dilation rate (d) within the DFDB. We applied the parameter d at three critical points: the first PSRB, the second PSRB, and the final 3×3 convolutional layer within the DFDB. According to the results presented in Table VII, modifying the value of d at the middle position enhances the reconstruction accuracy. Conversely, similar modifications to d at the tail position led to a decline in performance. This decrease in effectiveness may be attributed to the adjustments at the tail position, which interfere with the network's focus on crucial details, particularly in scenarios involving smaller objects. Tuning the dilation rate at this location appears to

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 09:18:05 UTC from IEEE Xplore. Restrictions apply.

TABLE V

QUANTITATIVE RESULTS FOR SR ON FIVE BENCHMARK DATASETS ARE PRESENTED. THE BEST RESULT IS HIGHLIGHTED IN RED, AND THE SECOND-BEST RESULT IS HIGHLIGHTED IN BLUE. THE MULTIADDS IS COMPUTED ON A 1280 × 720 HR IMAGE

Method	Scale	Params	M-Adds	Set5		Set14		BSD100		Urban100		Manga109	
				PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM
IDN [45]	×2	553K	124.6G	37.83	0.9600	33.30	0.9148	32.08	0.8985	31.27	0.9196	38.01	0.9749
CARN [44]		1592K	222.8G	37.76	0.9590	33.52	0.9166	32.09	0.8978	31.92	0.9256	38.36	0.9765
IMDN [31]		694K	158.8G	38.00	0.9605	33.63	0.9177	32.19	0.8996	32.17	0.9283	38.88	0.9774
RFDN [46]		534K	123.0G	38.05	0.9606	33.68	0.9184	32.16	0.8994	32.12	0.9278	38.88	0.9773
LMAN-S [54]		525K	119.9G	37.94	0.9603	33.49	0.9167	32.08	0.8984	31.85	0.9251	38.43	0.9765
SMSR [62]		985K	131.6G	38.00	0.9601	33.64	0.9179	32.17	0.8990	32.19	0.9284	38.76	0.9771
FDIWN-M [63]		433K	73.6G	38.03	0.9606	33.60	0.9179	32.17	0.8995	32.19	0.9284	-	-
VLESR [64]		311K	71.2G	38.01	0.9605	33.58	0.9177	32.16	0.8993	32.14	0.9280	38.75	0.9770
GASSL-S [65]		280K	65.5G	37.91	0.9602	33.53	0.9172	32.14	0.8992	31.81	0.9253	38.57	0.9769
FDSCSR-S [66]		466K	121.8G	38.02	0.9606	33.51	0.9174	32.18	0.8996	32.24	0.9288	38.67	0.9771
AMFFN(Ours)		298K	60.2G	38.07	0.9607	33.59	0.9178	32.21	0.9001	32.37	0.9299	38.89	0.9774
IDN [45]	×3	553K	56.3G	34.11	0.9253	29.99	0.8354	28.95	0.8013	27.42	0.8359	32.71	0.9381
CARN [44]		1592K	118.8G	34.29	0.9255	30.29	0.8407	29.06	0.8034	28.06	0.8493	33.50	0.9440
IMDN [31]		703K	71.5G	34.36	0.9270	30.32	0.8417	29.09	0.8046	28.17	0.8519	33.61	0.9445
RFDN [46]		541K	55.4G	34.41	0.9273	30.34	0.8420	29.09	0.8050	28.21	0.8525	33.67	0.9449
LMAN-S [54]		709K	73.1G	34.31	0.9265	30.24	0.8397	29.02	0.8030	28.02	0.8487	33.42	0.9433
SMSR [62]		993K	67.8G	34.40	0.9270	30.33	0.8412	29.10	0.8050	28.25	0.8536	33.68	0.9445
FDIWN-M [63]		446K	35.9G	34.46	0.9274	30.35	0.8423	29.10	0.8051	28.16	0.8528	-	-
VLESR [64]		319K	32.9G	34.40	0.9272	30.34	0.8415	29.08	0.8043	28.16	0.8519	33.61	0.9445
GASSL-S [65]		373K	39.4G	34.24	0.9260	30.28	0.8407	29.06	0.8038	27.95	0.8474	33.42	0.9434
FDSCSR-S [66]		471K	54.6G	34.42	0.9274	30.37	0.8429	29.10	0.8052	28.20	0.8532	33.55	0.9443
AMFFN(Ours)		305K	27.4G	34.48	0.9275	30.34	0.8420	29.11	0.8051	28.29	0.8544	33.72	0.9451
IDN [45]	×4	553K	32.3G	31.82	0.8903	28.25	0.7730	27.41	0.7297	25.41	0.7632	29.41	0.8942
CARN [44]		1592K	90.9G	32.13	0.8937	28.60	0.7806	27.58	0.7349	26.07	0.7837	30.47	0.9084
IMDN [31]		715K	40.9G	32.21	0.8948	28.58	0.7811	27.56	0.7353	26.04	0.7838	30.45	0.9075
RFDN [46]		550K	31.6G	32.24	0.8952	28.61	0.7819	27.57	0.7360	26.11	0.7858	30.58	0.9089
LMAN-S [54]		672K	65.1G	32.12	0.8939	28.53	0.7798	27.51	0.7340	25.96	0.7813	30.30	0.9062
SMSR [62]		1006K	41.6G	32.12	0.8932	28.55	0.7808	27.55	0.7351	26.11	0.7868	30.54	0.9085
FDIWN-M [63]		454K	19.6G	32.17	0.8941	28.55	0.7806	27.58	0.7364	26.02	0.7844	-	-
VLESR [64]		331K	19.5G	32.17	0.8945	28.55	0.7802	27.55	0.7345	26.03	0.7830	30.48	0.9073
GASSL-S [65]		428K	51.6G	32.01	0.8931	28.56	0.7808	27.56	0.7351	25.98	0.7818	30.35	0.9070
FDSCSR-S [66]		478K	31.1G	32.25	0.8959	28.61	0.7821	27.58	0.7367	26.12	0.7866	30.51	0.9087
AMFFN(Ours)		314K	15.9G	32.29	0.8958	28.62	0.7821	27.59	0.7365	26.22	0.7889	30.50	0.9083

adversely impact the network's ability to process this critical information effectively. Consequently, we ended up setting the default dilation rates as $d = 1, 2, 1$ to optimize performance.

2) Efficacy of the AMFF: To evaluate the effectiveness of the AMFF, we executed a comparative analysis using standard feature fusion as the principal architectural framework within the network. The comparison, as detailed in the first and third rows of Table VI, indicates enhanced performance achieved with a marginal increase in network parameters. This outcome provides substantial validation of the efficiency and effectiveness of our proposed AMFF structure.

To more comprehensively demonstrate the impact of AMFF, we made a visualization, as shown in Fig. 11. The LAM results clearly show that upon integrating AMFF, there is a marked intensification in the network's focus on vital information. This is evidenced by an increase in the DI value for "overpass63" from 4.59 to 5.70. Additionally, the SR results corroborate this, with the PSNR/SSIM values for the LR improving from 27.32/0.8561 to 27.47/0.8609 with AMFF. These advancements collectively validate the effectiveness of our proposed AMFF structure.

For a comprehensive assessment of the AMFF structure, we conducted a comparative analysis with alternative fusion structures, integrating each as the primary component of the network. These fusion structures are depicted in Fig. 6, with their corresponding performance evaluations detailed in Table VIII. Notably, the comparison with the AMFF structure reveals only a marginal change in performance when employing dense feature fusion. However, it is important to highlight that the network parameters increased from 314 to 383 K. In summary, our AMFF structure achieves a more effective balance between reconstruction accuracy and network parameters.

3) Influence of the Partial Ratio in PSConv: We explored the optimal hyperparameter settings for the partial ratio (r) within the PSConv. Various configurations of the parameter r were thoroughly investigated. Elevated values of r introduce a greater number of linear transformations. However, as indicated by the result in Table IX, we observed that the precision of reconstruction tends to plateau with the increment of r. Additionally, when we substituted PSConv with a standard 3×3 convolutional layer, there was a significant increase in network parameters accompanied by a noticeable decline in performance.

The increase in linear transformations complicates the optimization process within deep neural networks, leading to the observed phenomena of reduced reconstruction accuracy. Therefore, we opted for a partial ratio of $r = (1/3)$ to balance computational complexity and performance.

4) Influence of the Number of DFDBs: To optimize network parameters and performance, we examined how changing the number of DFDBs will affect them. The number of DFDBs,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 09:18:05 UTC from IEEE Xplore. Restrictions apply.

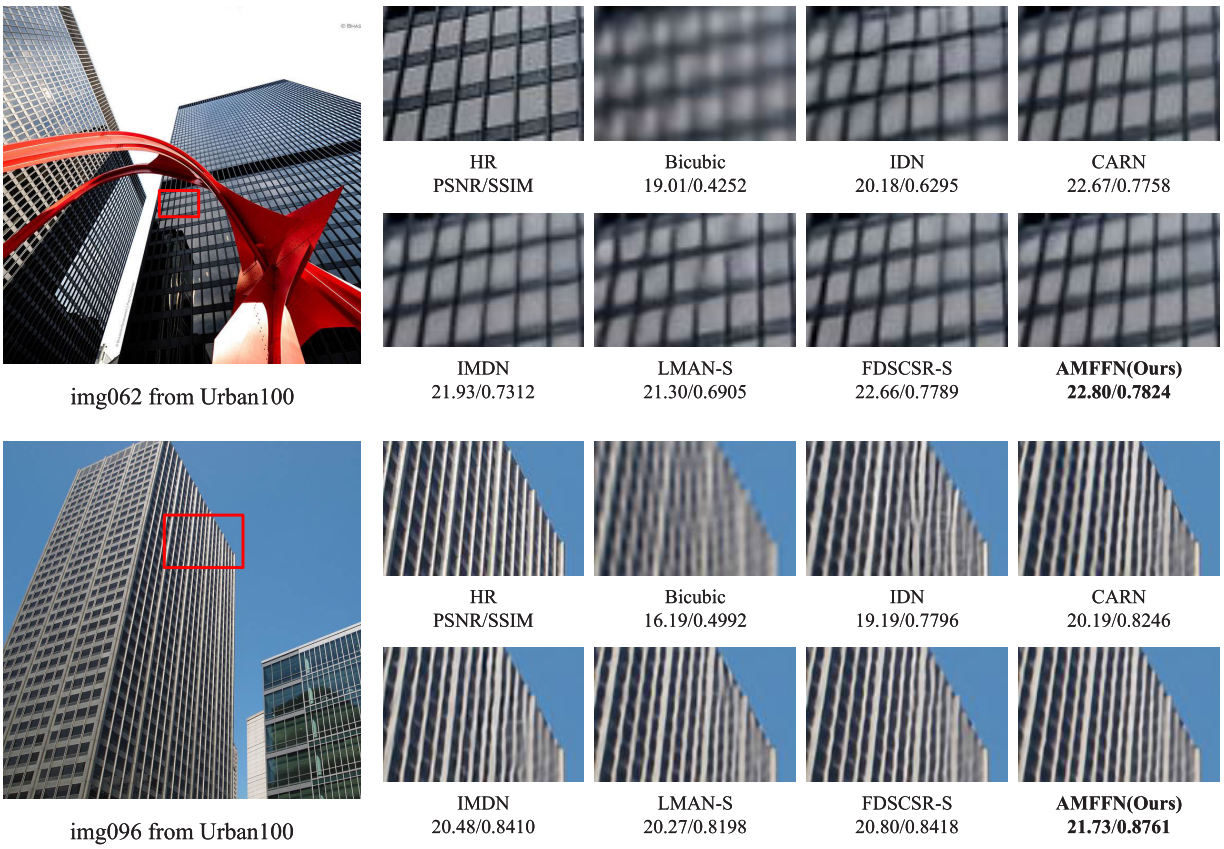


Fig. 10. Result comparison of our method with other lightweight methods on the Urban100 dataset (×4).

TABLE VI

ABLATION ANALYSIS FOR DFDB AND AMFF IN THE AMFFN NETWORK ON THE UC MERCED TEST DATASET FOR ×4 SR

Setting	DFDB	AMFF	Params	M-Adds	PSNR	SSIM
1	×	×	615K	33.4G	27.58	0.7624
2	✓	×	302K	15.4G	27.61	0.7628
3	×	✓	627K	34.0G	27.60	0.7628
4	✓	✓	314K	15.9G	**27.65**	**0.7641**

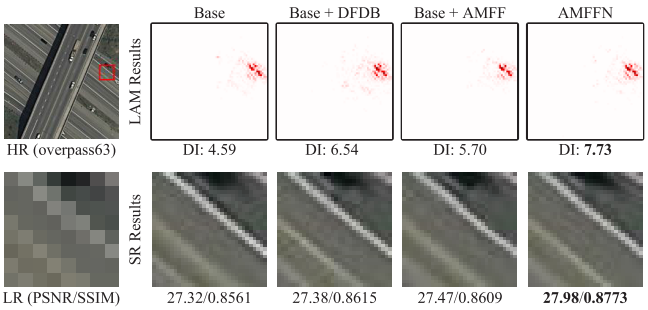


Fig. 11. Comparison of the SR results and LAM attribution results with different settings on UCMerced for ×4 SR.

a core component of the network, does influence the network's ultimate performance to a certain extent. Our consideration extends beyond performance alone, it also encompasses network parameters and multiadds operations. As indicated in Table X, there is a positive correlation between performance and incremental increases in n when n is within a lower range. This trend of enhanced performance persists until n reaches a critical point, identified as $n = 8$. Beyond this juncture, performance indicators have leveled off. Concurrently, network parameters, multiadds operations, and the depth of the network all increase with the rise in n. Since the main objective of this work is to examine the lightweight RSISR, when similar results are achieved, our preference is inclined toward selecting the setting with relatively fewer parameters and multiadds operations. Consequently, in this article, we adopt $n = 8$ as our final network configuration.

5) Influence of the Number of Network Channels: We examined how different network channel counts (e.g., 40, 48, 56, and 64) affected performance. The accuracy of the network is at its lowest when there are 40 channels, as seen by the data in Table XI. With the numerical value increases, the performance of our AMFFN can be further improved, suggesting its potential to achieve superior outcomes when implemented in large networks. However, it is essential to

TABLE VII

PERFORMANCE RESULTS WITH DIFFERENT SETTINGS OF d ON THE UC MERCED TEST DATASET FOR ×4 SR

Variant	PSNR	SSIM
d = 1, 1, 1	27.64	0.7639
d = 1, 2, 1	**27.65**	**0.7641**
d = 1, 2, 3	27.57	0.7622
d = 1, 2, 5	27.56	0.7619

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 09:18:05 UTC from IEEE Xplore. Restrictions apply.

TABLE VIII
PERFORMANCE RESULTS WITH DIFFERENT FUSION STRUCTURES ON THE UC MERCED TEST DATASET FOR ×4 SR

Variant	Params	M-Adds	PSNR	SSIM
a	302K	15.4G	27.61	0.7628
b	383K	20.0G	**27.66**	**0.7653**
c	295K	15.0G	27.62	0.7632
d	314K	15.9G	27.65	0.7641

TABLE IX
PERFORMANCE RESULTS WITH DIFFERENT SETTINGS OF r ON THE UC MERCED TEST DATASET FOR ×4 SR

Variant	Params	M-Adds	PSNR	SSIM
r = 1/4	296K	14.9G	27.62	0.7632
r = 1/3	314K	15.9G	**27.65**	0.7641
r = 1/2	363K	18.8G	**27.65**	**0.7647**
w/ 3×3Conv	588K	31.7G	27.63	0.7637

TABLE X
PERFORMANCE RESULTS WITH DIFFERENT SETTINGS OF n ON THE UC MERCED TEST DATASET FOR ×4 SR

Variant	Params	M-Adds	PSNR	SSIM
n = 2	99K	5.1G	27.33	0.7529
n = 4	170K	8.7G	27.53	0.7607
n = 6	242K	12.3G	27.61	0.7628
n = 8	314K	15.9G	**27.65**	0.7641
n = 10	385K	19.5G	**27.65**	**0.7650**
n = 12	457K	23.1G	**27.65**	0.7643

TABLE XI
PERFORMANCE RESULTS WITH DIFFERENT SETTINGS OF c ON THE UC MERCED TEST DATASET FOR ×4 SR

Variant	Params	M-Adds	PSNR	SSIM
c = 40	219K	11.1G	27.58	0.7628
c = 48	314K	15.9G	27.65	0.7641
c = 56	414K	21.1G	27.68	0.7654
c = 64	541K	27.6G	**27.70**	**0.7666**

note that the number of parameters and multiadds operations also increases proportionally. To achieve a lightweight RSISR, we have determined that the optimal number of network channels is 48.

V. CONCLUSION

In this article, we propose a lightweight RSISR network called AMFFN for reconstructing LR images with limited time and space overhead. Specifically, to reduce the redundant convolution operation, we invoke PConv to design a more efficient feature extraction block PSRB. Meanwhile, we design a novel distillation structure DFDB based on PSRB and the adjustable dilation rate to enhance the adaptability and versatility of the network. Furthermore, we have meticulously crafted the AMFF structure, which utilizes a grouping method for the progressive fusion of multilevel features, culminating in a comprehensive feature fusion representation. The obtained fused features are finally fed into the reconstruction module for generating HR images. Extensive experiments on two RSI datasets and five natural image benchmarks show that our method achieves significant advantages in balancing computational complexity and performance.

However, the scope of this article is confined to RSISR with a bicubic downsampler, and it does not venture into other domains such as blind RSISR, continuous RSISR, or hyperspectral image SR. Future investigations will aim to probe the synergy between our model and additional techniques, thereby broadening the applicability of our model to encompass a more diverse range of RSISR challenges.

REFERENCES

[1] K. Li, G. Wan, G. Cheng, L. Meng, and J. Han, "Object detection in optical remote sensing images: A survey and a new benchmark," *ISPRS J. Photogramm. Remote Sens.*, vol. 159, pp. 296–307, Jan. 2020.

[2] A. M. Lechner, G. M. Foody, and D. S. Boyd, "Applications in remote sensing to forest ecology and management," *One Earth*, vol. 2, no. 5, pp. 405–412, May 2020.

[3] J. Wang, Y. Zhong, Z. Zheng, A. Ma, and L. Zhang, "RSNet: The search for remote sensing deep neural networks in recognition tasks," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 3, pp. 2520–2534, Mar. 2021.

[4] X. Zhang et al., "Remote sensing image super-resolution via dual-resolution network based on connected attention mechanism," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5611013.

[5] J. Wang, B. Wang, X. Wang, Y. Zhao, and T. Long, "Hybrid attention-based U-shaped network for remote sensing image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5612515.

[6] Q. Li, M. Gong, Y. Yuan, and Q. Wang, "Symmetrical feature propagation network for hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5536912.

[7] P. Wang, B. Bayram, and E. Sertel, "A comprehensive review on deep learning based remote sensing image super-resolution methods," *Earth-Sci. Rev.*, vol. 232, Sep. 2022, Art. no. 104110.

[8] J. Li et al., "Deep learning in multimodal remote sensing data fusion: A comprehensive review," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 112, Aug. 2022, Art. no. 102926.

[9] M. Hu, K. Jiang, Z. Wang, X. Bai, and R. Hu, "CycMuNet+: Cycle-projected mutual learning for spatial–temporal video super-resolution," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 11, pp. 13376–13392, Nov. 2023.

[10] X. Zhang, S. Cheng, L. Wang, and H. Li, "Asymmetric cross-attention hierarchical network based on CNN and transformer for bitemporal remote sensing images change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 2000415.

[11] Z. Chao, S. Cheng, and Y. Li, "Deep internally connected transformer hashing for image retrieval," *Knowledge-Based Syst.*, vol. 279, Nov. 2023, Art. no. 110953.

[12] K. Zheng et al., "Coupled convolutional neural network with adaptive response function learning for unsupervised hyperspectral super resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 3, pp. 2487–2502, Mar. 2021.

[13] L. Gao, J. Li, K. Zheng, and X. Jia, "Enhanced autoencoders with attention-embedded degradation learning for unsupervised hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5509417.

[14] J. Li, K. Zheng, Z. Li, L. Gao, and X. Jia, "X-shaped interactive autoencoders with cross-modality mutual learning for unsupervised hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5518317.

[15] Y. Xiao, Q. Yuan, K. Jiang, J. He, Y. Wang, and L. Zhang, "From degrade to upgrade: Learning a self-supervised degradation guided adaptive network for blind remote sensing image super-resolution," *Inf. Fusion*, vol. 96, pp. 297–311, Aug. 2023.

[16] M. Qin et al., "Remote sensing single-image resolution improvement using a deep gradient-aware network with image-specific enhancement," *Remote Sens.*, vol. 12, no. 5, p. 758, Feb. 2020.

[17] Q. Li, Y. Yuan, X. Jia, and Q. Wang, "Dual-stage approach toward hyperspectral image super-resolution," *IEEE Trans. Image Process.*, vol. 31, pp. 7252–7263, 2022.

[18] R. Dong, L. Mou, L. Zhang, H. Fu, and X. X. Zhu, "Real-world remote sensing image super-resolution via a practical degradation model and a kernel-aware network," *ISPRS J. Photogramm. Remote Sens.*, vol. 191, pp. 155–170, Sep. 2022.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 09:18:05 UTC from IEEE Xplore. Restrictions apply.

[19] D. Zhang, J. Shao, X. Li, and H. T. Shen, "Remote sensing image super-resolution via mixed high-order attention network," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 6, pp. 5183–5196, Jun. 2021.

[20] S. Lei, Z. Shi, and W. Mo, "Transformer-based multistage enhancement for remote sensing image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5615611.

[21] J. Tu, G. Mei, Z. Ma, and F. Piccialli, "SWCGAN: Generative adversarial network combining Swin transformer and CNN for remote sensing image super-resolution," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 5662–5673, 2022.

[22] K. Jiang, Z. Wang, P. Yi, G. Wang, T. Lu, and J. Jiang, "Edge-enhanced GAN for remote sensing image superresolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 8, pp. 5799–5812, Aug. 2019.

[23] Q. Li, M. Gong, Y. Yuan, and Q. Wang, "RGB-induced feature modulation network for hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5512611.

[24] Y. Xiao, Q. Yuan, K. Jiang, J. He, X. Jin, and L. Zhang, "EDiffSR: An efficient diffusion probabilistic model for remote sensing image super-resolution," 2023, *arXiv:2310.19288*.

[25] S. Wang, T. Zhou, Y. Lu, and H. Di, "Contextual transformation network for lightweight remote-sensing image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5615313.

[26] Z. Wang et al., "FeNet: Feature enhancement network for lightweight remote-sensing image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5622112.

[27] J. Chen et al., "Run, don't walk: Chasing higher flops for faster neural networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 12021–12031.

[28] J. Li, F. Fang, K. Mei, and G. Zhang, "Multi-scale residual network for image super-resolution," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 517–532.

[29] F. Liu, X. Yang, and B. De Baets, "Lightweight image super-resolution with a feature-refined network," *Signal Process., Image Commun.*, vol. 111, Feb. 2023, Art. no. 116898.

[30] X. Luo, Y. Xie, Y. Zhang, Y. Qu, C. Li, and Y. Fu, "LatticeNet: Towards lightweight image super-resolution with lattice block," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2020, pp. 272–289.

[31] Z. Hui, X. Gao, Y. Yang, and X. Wang, "Lightweight image super-resolution with information multi-distillation network," in *Proc. 27th ACM Int. Conf. Multimedia*, Oct. 2019, pp. 2024–2032.

[32] C. Dong, C. C. Loy, K. He, and X. Tang, "Image super-resolution using deep convolutional networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 38, no. 2, pp. 295–307, Feb. 2016.

[33] C. Dong, C. C. Loy, and X. Tang, "Accelerating the super-resolution convolutional neural network," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2016, pp. 391–407.

[34] J. Kim, J. K. Lee, and K. M. Lee, "Accurate image super-resolution using very deep convolutional networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 1646–1654.

[35] Y. Zhang, K. Li, K. Li, L. Wang, B. Zhong, and Y. Fu, "Image super-resolution using very deep residual channel attention networks," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 286–301.

[36] J. Liu, W. Zhang, Y. Tang, J. Tang, and G. Wu, "Residual feature aggregation network for image super-resolution," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 2356–2365.

[37] J. Liang, J. Cao, G. Sun, K. Zhang, L. Van Gool, and R. Timofte, "SwinIR: Image restoration using Swin transformer," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. Workshops (ICCVW)*, Oct. 2021, pp. 1833–1844.

[38] X. Chen, X. Wang, J. Zhou, Y. Qiao, and C. Dong, "Activating more pixels in image super-resolution transformer," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 22367–22377.

[39] S. Lei, Z. Shi, and Z. Zou, "Super-resolution for remote sensing images via local–global combined network," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 8, pp. 1243–1247, Aug. 2017.

[40] J. M. Haut, M. E. Paoletti, R. Fernandez-Beltran, J. Plaza, A. Plaza, and J. Li, "Remote sensing single-image superresolution based on a deep compendium model," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 9, pp. 1432–1436, Sep. 2019.

[41] X. Dong, X. Sun, X. Jia, Z. Xi, L. Gao, and B. Zhang, "Remote sensing image super-resolution using novel dense-sampling networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 2, pp. 1618–1633, Feb. 2021.

[42] H. Wu, N. Ni, and L. Zhang, "Learning dynamic scale awareness and global implicit functions for continuous-scale super-resolution of remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5602315.

[43] K. Chen et al., "Continuous remote sensing image super-resolution based on context interaction in implicit function space," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4702216.

[44] N. Ahn, B. Kang, and K.-A. Sohn, "Fast, accurate, and lightweight super-resolution with cascading residual network," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 252–268.

[45] Z. Hui, X. Wang, and X. Gao, "Fast and accurate single image super-resolution via information distillation network," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 723–731.

[46] J. Liu, J. Tang, and G. Wu, "Residual feature distillation network for lightweight image super-resolution," in *Proc. Eur. Conf. Comput. Vis. Workshop (ECCVW)*, 2020, pp. 41–55.

[47] F. Kong et al., "Residual local feature network for efficient super-resolution," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2022, pp. 765–775.

[48] D. Hendrycks and K. Gimpel, "Gaussian error linear units (GELUs)," 2016, *arXiv:1606.08415*.

[49] Y. Yang and S. Newsam, "Bag-of-visual-words and spatial extensions for land-use classification," in *Proc. 18th SIGSPATIAL Int. Conf. Adv. Geographic Inf. Syst.*, Nov. 2010, pp. 270–279.

[50] Q. Zou, L. Ni, T. Zhang, and Q. Wang, "Deep learning based feature selection for remote sensing scene classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 12, no. 11, pp. 2321–2325, Nov. 2015.

[51] Z. Wang, A. C. Bovik, H. R. Sheikh, and E. P. Simoncelli, "Image quality assessment: From error visibility to structural similarity," *IEEE Trans. Image Process.*, vol. 13, no. 4, pp. 600–612, Apr. 2004.

[52] C. Tian et al., "Lightweight image super-resolution with enhanced CNN," *Knowl.-Based Syst.*, vol. 205, Oct. 2020, Art. no. 106235.

[53] C. Tian, Y. Xu, W. Zuo, B. Zhang, L. Fei, and C.-W. Lin, "Coarse-to-fine CNN for image super-resolution," *IEEE Trans. Multimedia*, vol. 23, pp. 1489–1502, 2021.

[54] J. Wan, H. Yin, Z. Liu, A. Chong, and Y. Liu, "Lightweight image super-resolution by multi-scale aggregation," *IEEE Trans. Broadcast.*, vol. 67, no. 2, pp. 372–382, Jun. 2021.

[55] W. Deng, H. Yuan, L. Deng, and Z. Lu, "Reparameterized residual feature network for lightweight image super-resolution," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2023, pp. 1712–1721.

[56] E. Agustsson and R. Timofte, "NTIRE 2017 challenge on single image super-resolution: Dataset and study," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jul. 2017, pp. 1122–1131.

[57] M. Bevilacqua, A. Roumy, C. Guillemot, and M.-L.-A. Morel, "Low-complexity single-image super-resolution based on nonnegative neighbor embedding," in *Proc. Brit. Mach. Vis. Conf.*, 2012, pp. 1–10.

[58] J. Yang, J. Wright, T. S. Huang, and Y. Ma, "Image super-resolution via sparse representation," *IEEE Trans. Image Process.*, vol. 19, no. 11, pp. 2861–2873, Nov. 2010.

[59] D. Martin, C. Fowlkes, D. Tal, and J. Malik, "A database of human segmented natural images and its application to evaluating segmentation algorithms and measuring ecological statistics," in *Proc. 8th IEEE Int. Conf. Comput. Vis. (ICCV)*, Jul. 2001, pp. 416–423.

[60] J.-B. Huang, A. Singh, and N. Ahuja, "Single image super-resolution from transformed self-exemplars," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 5197–5206.

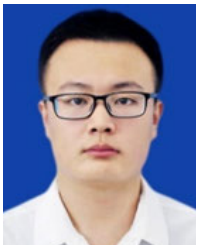
[61] Y. Matsui et al., "Sketch-based Manga retrieval using Manga109 dataset," *Multimedia Tools Appl.*, vol. 76, no. 20, pp. 21811–21838, Oct. 2017.

[62] L. Wang et al., "Exploring sparsity in image super-resolution for efficient inference," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 4915–4924.

[63] G. Gao, W. Li, J. Li, F. Wu, H. Lu, and Y. Yu, "Feature distillation interaction weighting network for lightweight image super-resolution," in *Proc. AAAI Conf. Artif. Intell.*, 2022, vol. 36, no. 1, pp. 661–669.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 09:18:05 UTC from IEEE Xplore. Restrictions apply.

[64] D. Gao and D. Zhou, "A very lightweight and efficient image super-resolution network," *Exp. Syst. Appl.*, vol. 213, Mar. 2023, Art. no. 118898.
[65] H. Wang, Y. Zhang, C. Qin, L. Van Gool, and Y. Fu, "Global aligned structured sparsity learning for efficient image super-resolution," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 9, pp. 10974–10989, Sep. 2023.
[66] Z. Wang, G. Gao, J. Li, H. Yan, H. Zheng, and H. Lu, "Lightweight feature de-redundancy and self-calibration network for efficient image super-resolution," *ACM Trans. Multimedia Comput., Commun., Appl.*, vol. 19, no. 3, pp. 1–15, Aug. 2023.
[67] J. Gu and C. Dong, "Interpreting super-resolution networks with local attribution maps," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 9195–9204.



Hongyuan Wang received the B.S. degree from the School of Information and Engineering, Ocean University of China, Qingdao, China, in 2022. He is currently pursuing the M.S. degree with Xinjiang University, Ürümqi, China.

His research interests include computer vision and image super-resolution.



Shuli Cheng received the Ph.D. degree from the School of Information and Engineering, Xinjiang University, Ürümqi, China, in 2021.

He is currently working as an Associate Professor at the School of Computer Science and Technology, Xinjiang University. His research interests include computer vision, natural language processing, multimodal processing, and remote-sensing information processing.



Yongming Li received the master's degree from the School of Information and Engineering, Xinjiang University, Ürümqi, China, in 2007.

He is now an Associate Professor at the School of Information Science and Engineering, Xinjiang University. His research interests include computer vision, natural language processing, and wireless sensor networks.



Anyu Du received the master's degree from the School of Information and Engineering, Xinjiang University, Ürümqi, China, in 2019.

She has a deep interest in the wireless sensor network and digital image processing. She is also doing some research in multifeature fusion image retrieval.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 09:18:05 UTC from IEEE Xplore. Restrictions apply.